Three quadratic equations for two translated elliptic parameters

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Abstract

Let $E : y^2 = x^3 - Dx$ over a field of characteristic zero and let $P = (a, b)$ be a fixed point satisfying the specified nonvanishing conditions. We give three quadratic equations in four field coordinates whose solutions describe exactly the points H for which H and $H - P$ are finite with nonzero abscissas. The coordinates retain both reciprocal abscissas and both values of y/x. All exceptional leading coefficients are treated explicitly. The resulting direct equations for formal-parameter ratios save one common field coordinate and one field coordinate for each numerator pair. Every identity has a separate exact verification using only the Python standard library.

1 The quadratic equations

Let k be a field of characteristic zero, let $D \in k^\times$, and put

$$E : y^2 = x^3 - Dx, \qquad P = (a, b) \in E(k).$$

Assume

$$ab \neq 0, \qquad C := D^2 + 6a^2D - 3a^4 \neq 0. \tag{1}$$

Write $A = a^2 + D$ and $B = a^2 - D$. The point equation is $b^2 = aB$. For four field coordinates S, T, f, g, define

$$\begin{aligned} L_f &= 2bD(aT - b), & Q_f &= aAST + bAS - bBT + 4a^2D, \\ L_g &= 2bD(aS + b), & Q_g &= -aAST - bBS + bAT - 4a^2D. \end{aligned}$$

Consider the three quadratic equations

$$L_f f = Q_f, \qquad L_g g = Q_g, \qquad D^2fg + aD(f + g) + aST + b(S - T) + D = 0. \tag{2}$$

No inverse witness occurs in this system.

Theorem 1.1. *Equations (2) describe exactly the points*

$$H \in E(k) \setminus \{O, (0, 0), P, P + (0, 0)\}$$

through the identities

$$S = \frac{y(H)}{x(H)}, \quad T = \frac{y(H - P)}{x(H - P)}, \quad f = \frac{1}{x(H)}, \quad g = \frac{1}{x(H - P)}. \tag{3}$$

In particular every solution has $fg \neq 0$. Every allowed point has one and only one quadruple of coordinates satisfying these identities, and every solution of the quadratic system is such a quadruple. The theorem also holds when $a^2 + D = 0$.

2 Polynomial reconstruction and its exceptional cases

Define

$$\Phi = aS^2T^2 - B(S^2+T^2) + 2AST + 4aD$$

and two polynomial expressions in S, T:

$$\begin{aligned} 2bDf_0 &= -AS + BT - bS^2 - aS^2T, \\ 2bDg_0 &= AT - BS - bT^2 + aST^2. \end{aligned} \tag{4}$$

The divisions here are only by the fixed nonzero constant $2bD$. The following identities hold as polynomial identities modulo $b^2 - a^3 + aD$:

$$Df_0^2 + S^2 f_0 - 1 = \frac{a^2S^2 - b^2}{4ab^2D}\Phi, \tag{5}$$

$$Dg_0^2 + T^2 g_0 - 1 = \frac{a^2T^2 - b^2}{4ab^2D}\Phi,$$

$$L_f f_0 - Q_f = -a\Phi, \qquad L_g g_0 - Q_g = a\Phi. \tag{6}$$

Set

$$r_f = L_f f - Q_f, \qquad r_g = L_g g - Q_g, \qquad R = D^2 fg + aD(f+g) + aST + b(S-T) + D.$$

The elimination identity is

$$\begin{aligned} L_f L_g R = {} & D^2 r_f r_g + (aDL_g + D^2 Q_g) r_f \\ & + (aDL_f + D^2 Q_f) r_g - aD^2 C\Phi. \end{aligned} \tag{7}$$

One also has

$$(2bD)^2 R(f_0, g_0) = D^2\big(B - aST - b(S-T)\big)\Phi. \tag{8}$$

All these identities follow by expansion and substitution of $b^2 = a(a^2 - D)$. In particular, no division by L_f or L_g is used in (7).

Lemma 2.1. *Every solution of (2) satisfies $\Phi = 0$ and $f = f_0$, $g = g_0$.*

Proof. The nonvanishing of aD^2C and (7) give $\Phi = 0$. When $L_f L_g \neq 0$, equations (6) give the claim. Suppose $L_f = 0$. Then $T = b/a$. If $A = 0$, substitution gives $\Phi = -8a^3 \neq 0$, a contradiction. If $A \neq 0$, the equation $\Phi = 0$ is linear in S and gives

$$S = \frac{a^4 - 6Da^2 + D^2}{2bA}, \qquad L_g = -\frac{aDC}{A} \neq 0.$$

Thus $g = g_0 = 1/a$. The coefficient of f in R is $D^2/a + aD = DA/a \neq 0$, and (8) then gives $f = f_0$. Interchanging S and T and replacing b by $-b$ handles $L_g = 0$. The two leading coefficients cannot vanish together: at $S = -b/a, T = b/a$, one has $Q_f = C \neq 0$. □

3 Identification with elliptic translation

By (5), the equation $\Phi = 0$ implies $f_0 g_0 \neq 0$. It therefore supplies points

$$H = (f_0^{-1}, S/f_0), \qquad H' = (g_0^{-1}, T/g_0)$$

on E. Two further identities are

$$a(f_0 T + g_0 S) + b(g_0 - f_0) = S + T \tag{9}$$

and

$$f_0 g_0 (aS+b)^2 - a(1-af_0)^2 = -\frac{(aS+b)^2(aST+b(S-T)+D)}{4b^2D^2}\Phi. \tag{10}$$

The first is an identity even before imposing $\Phi = 0$.

If $x = x(H) \neq a$, the slope of the chord through H and $-P$ is

$$\mu = \frac{xS+b}{x-a}.$$

The curve equation $x^2 - S^2x - D = 0$ and $b^2 = a(a^2 - D)$ give

$$\mu^2 - x - a = \frac{x(aS+b)^2}{a(x-a)^2}.$$

Equation (10) says that this is $g_0^{-1} = x(H')$. After multiplying by $f_0 g_0$, equation (9) is precisely the collinearity condition for $H, -H', -P$. It therefore identifies the ordinate too, and proves $H' = H - P$.

Suppose $x(H) = a$. Then $S = \pm b/a$. At $S = b/a$ the expression for f_0 is $-a/D$. If $A \neq 0$, this differs from $1/a$, so the case is impossible. If $A = 0$, substitution in Φ gives $-8a^3 \neq 0$, which is again impossible. At $S = -b/a$, the same $A = 0$ calculation excludes the case; otherwise

$$f_0 = 1/a, \quad T = -\frac{a^4 - 6Da^2 + D^2}{2bA}, \quad g_0 = \frac{4b^2}{A^2}.$$

The doubling formulas identify $H = -P$ and $H' = -2P$. This proves the translation assertion for every solution.

Conversely, let H and $H - P$ be finite with nonzero abscissas. For $x(H) \neq a$, substitution of the chord formulas gives $\Phi = 0$ and (4). The preceding explicit doubling calculation covers $H = -P$; the other possibility $H = P$ is outside the allowed set. Equations (6) and (8) then imply (2). Uniqueness follows from the four rational functions in (3). This proves Theorem 1.1.

4 Necessity of an exceptional-constant condition

The condition $C \neq 0$ cannot simply be dropped. Over $k = \mathbb{Q}(\sqrt{3})$, take

$$a = 1, \qquad D = -3 + 2\sqrt{3}, \qquad b = \sqrt{3} - 1.$$

Then $abD \neq 0$, $b^2 = 1 - D$, and $C = 0$. Set

$$S = -b, \qquad T = b, \qquad f = 0, \qquad g = 3/D - 4.$$

Here $L_f = L_g = Q_f = Q_g = 0$, while the third equation reduces to $Dg + 4D - 3 = 0$. Thus this is a solution of the three quadrics with $f = 0$, and it cannot have the interpretation (3). The elimination identity identifies the precise constant responsible for this failure.

5 Direct equations for formal-parameter ratios

Write $z = -x/y$. Suppose numerator points have a quadratic description with coordinates V, Q, I, K such that

$$z(U) = -V/I, \qquad z(V_1) = -Q/K, \qquad IK \neq 0.$$

Zero numerators may represent the identity. At a common-point solution with $ST \neq 0$, the two ratios are given directly by

$$LI = VS, \qquad L'K = QT.$$

These are quadratic equations. They have exactly the values $L = z(U)/z(H)$ and $L' = z(V_1)/z(H - P)$, including either identity numerator. Consequently no separate coordinate for $z(V_1)$ is needed.

Over a number field of degree d, expansion in a fixed rational basis turns each field coordinate into d rational coordinates and each quadratic field equation into d rational quadratic equations. The common-point model uses $4d$ coordinates and $3d$ equations. In a construction with r numerator pairs, replacing a five-coordinate common chart and eliminating one numerator parameter per pair saves $d(r + 1)$ rational coordinates, provided the construction already supplies the stated domains and the numerator identities. This is an algebraic statement about representations; it does not supply any Diophantine definition of a ring of integers by itself.

For the applications over $\mathbb{Q}(\rho)$, $\rho^2 = 2$, the point is

$$a = \frac{57 - 28\rho}{8}, \qquad b = \frac{-660 + 417\rho}{32}, \qquad D = 11 - 6\rho.$$

Exact calculation gives

$$C = -\frac{94897379}{4096} + \frac{4195821}{256}\rho, \qquad \mathrm{N}(C) = -\frac{8203357319351}{16777216} \neq 0.$$

Thus the common-chart saving is two rational coordinates, and each numerator pair saves two more. All other arithmetic arguments remain those of the corresponding defining-polynomial construction.

6 Exact verification

The program

`computation/elliptic_constraints/check_fixed_difference_chart.py`

verifies twelve general identities in the quotient ring

$$\mathbb{Z}[a, b, D, S, T]/(b^2 - a^3 + aD).$$

It uses explicit integer polynomial arithmetic, imports no producer, and checks the master identity without division. It also checks the curve identities, translation, reconstruction, and the nonzero coefficients in the exceptional cases. Independent rational-point checks compare the formulas with the chord-and-tangent law at $H = -P, 2P, 3P$ on $y^2 = x^3 - 2x$ with $P = (2, 2)$. Its output is `fixed-difference-chart-certificate.json`. The separate quartic checker verifies the displayed nonzero constant in $\mathbb{Q}(\sqrt{2})$ and the resulting rational component equations. The identities prove the assertions for every allowed field and point; the sample points serve only as additional verification.